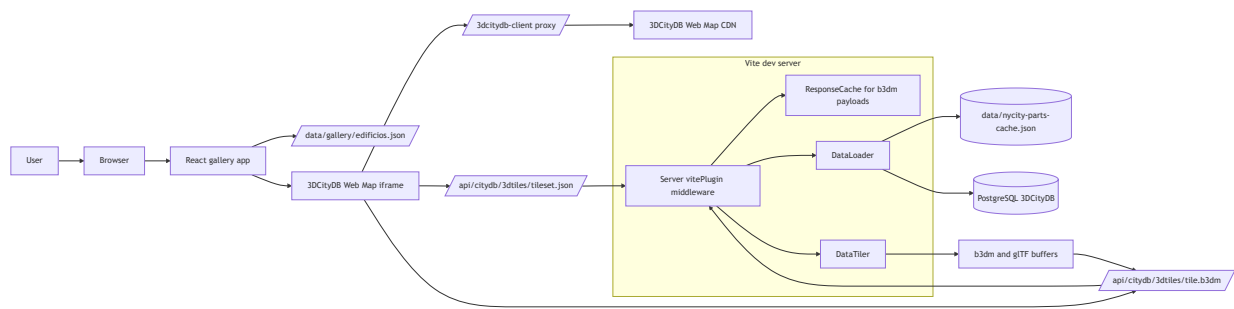


Architecture diagrams

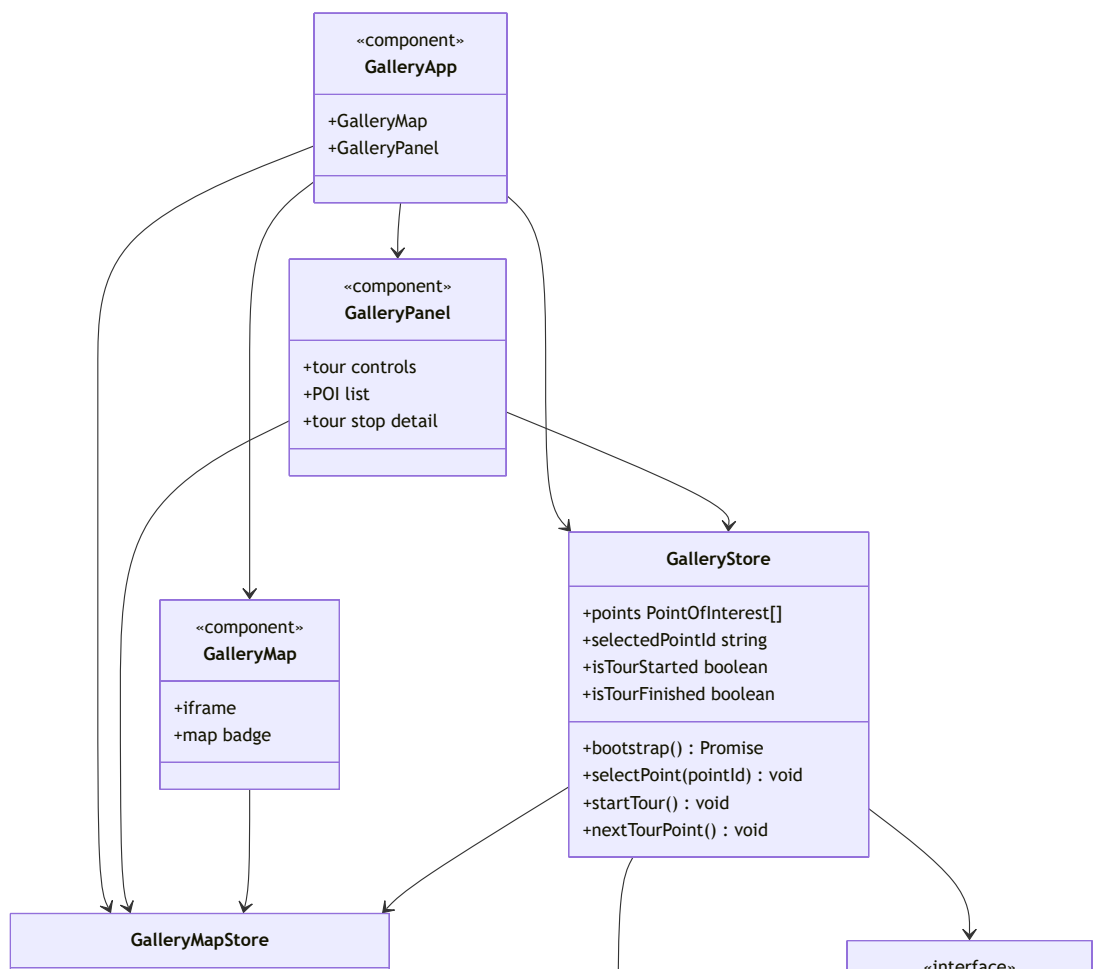
This document summarizes the most important runtime relationships in the CityGML LoD viewer. The diagrams are written in Mermaid so they can be rendered directly by GitHub, VS Code extensions, or Mermaid CLI.

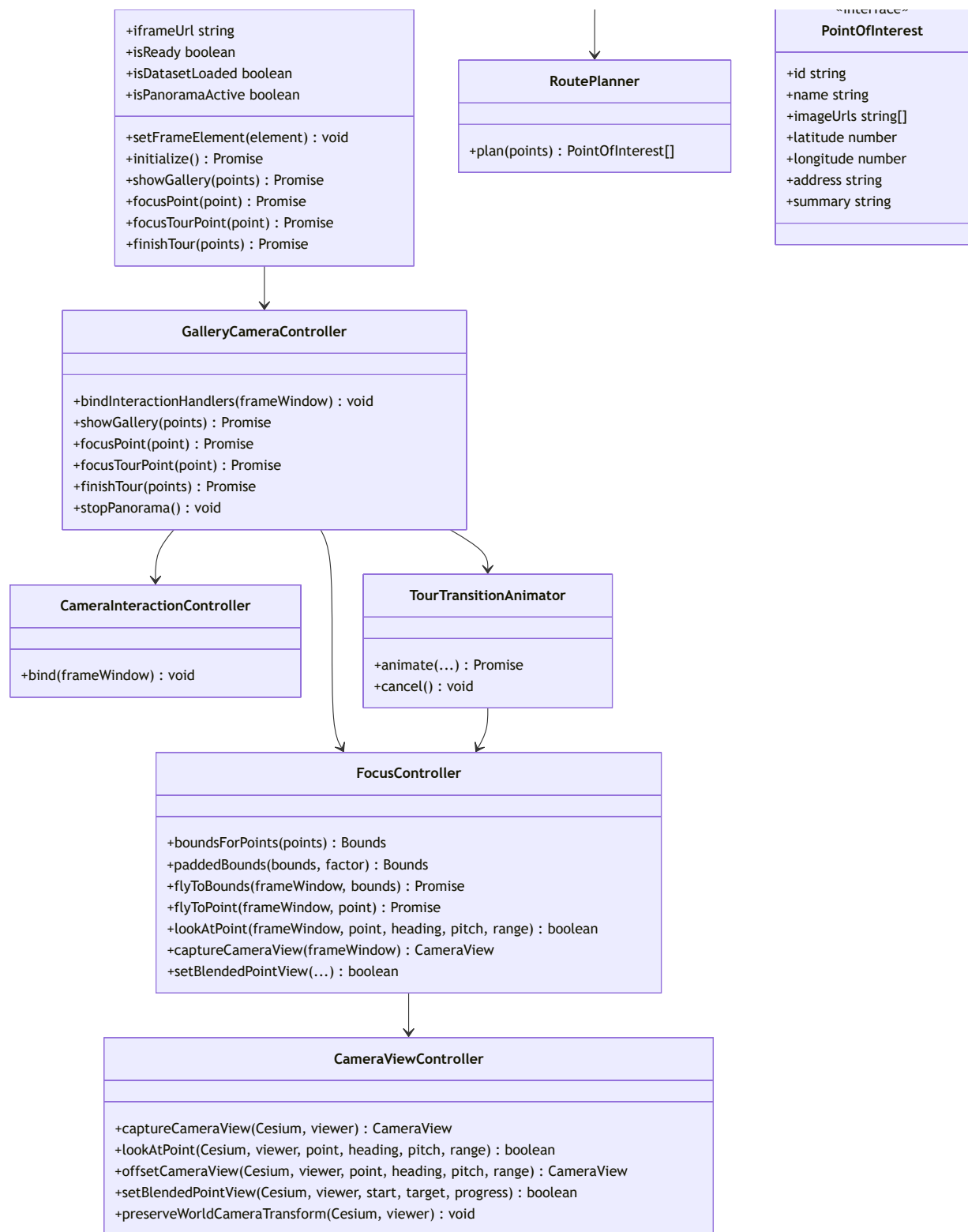
System communication



The browser hosts two UI layers: the local React/MobX gallery and the proxied 3DCityDB Web Map client inside an iframe. The iframe loads Cesium 3D Tiles from the local Vite middleware, while gallery points are loaded from static JSON and rendered into the embedded Cesium viewer by [GalleryMapStore](#).

Client classes



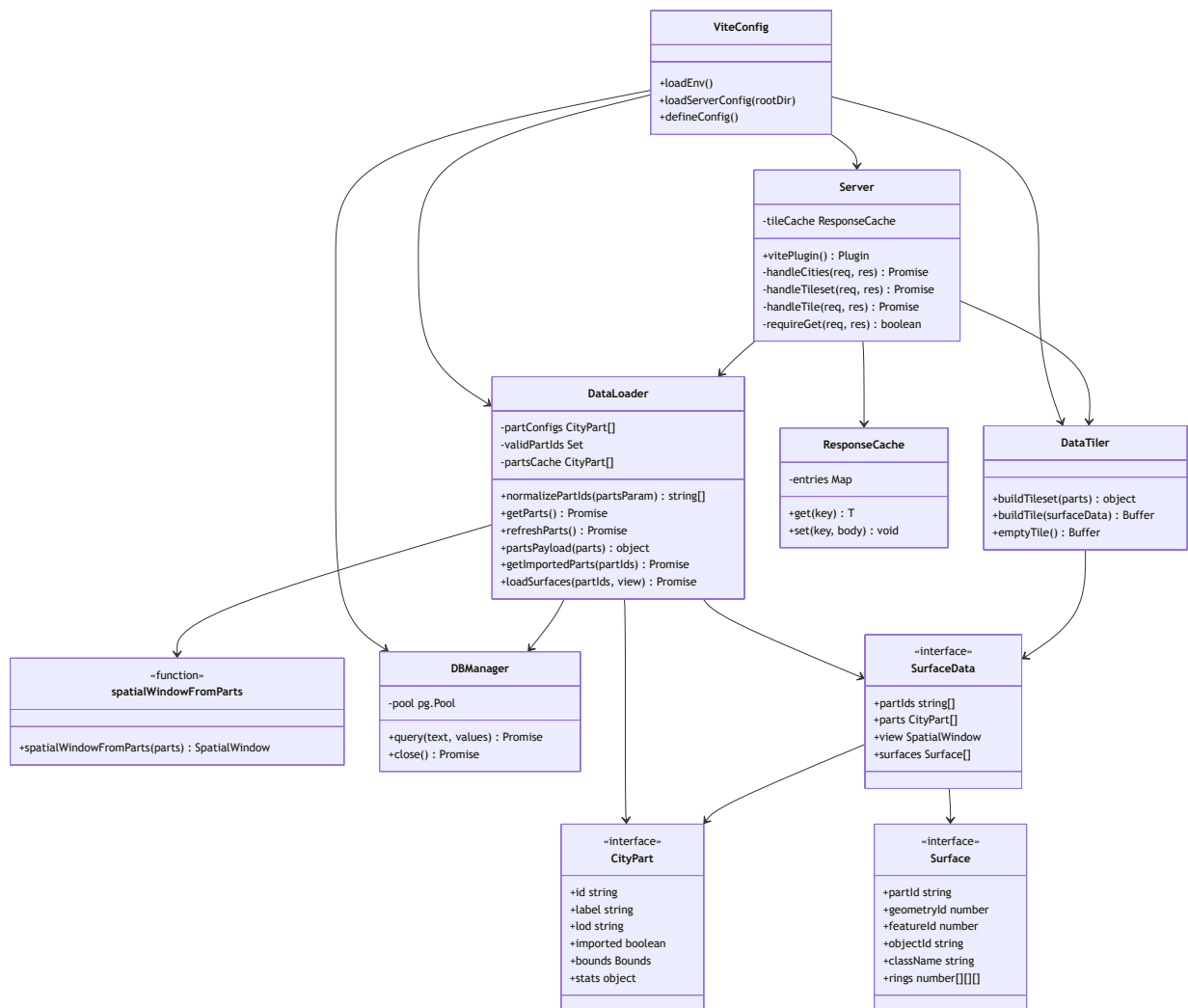


Relevant source files:

- [src/client/main.tsx](#)
- [src/client/app/GalleryApp.tsx](#)

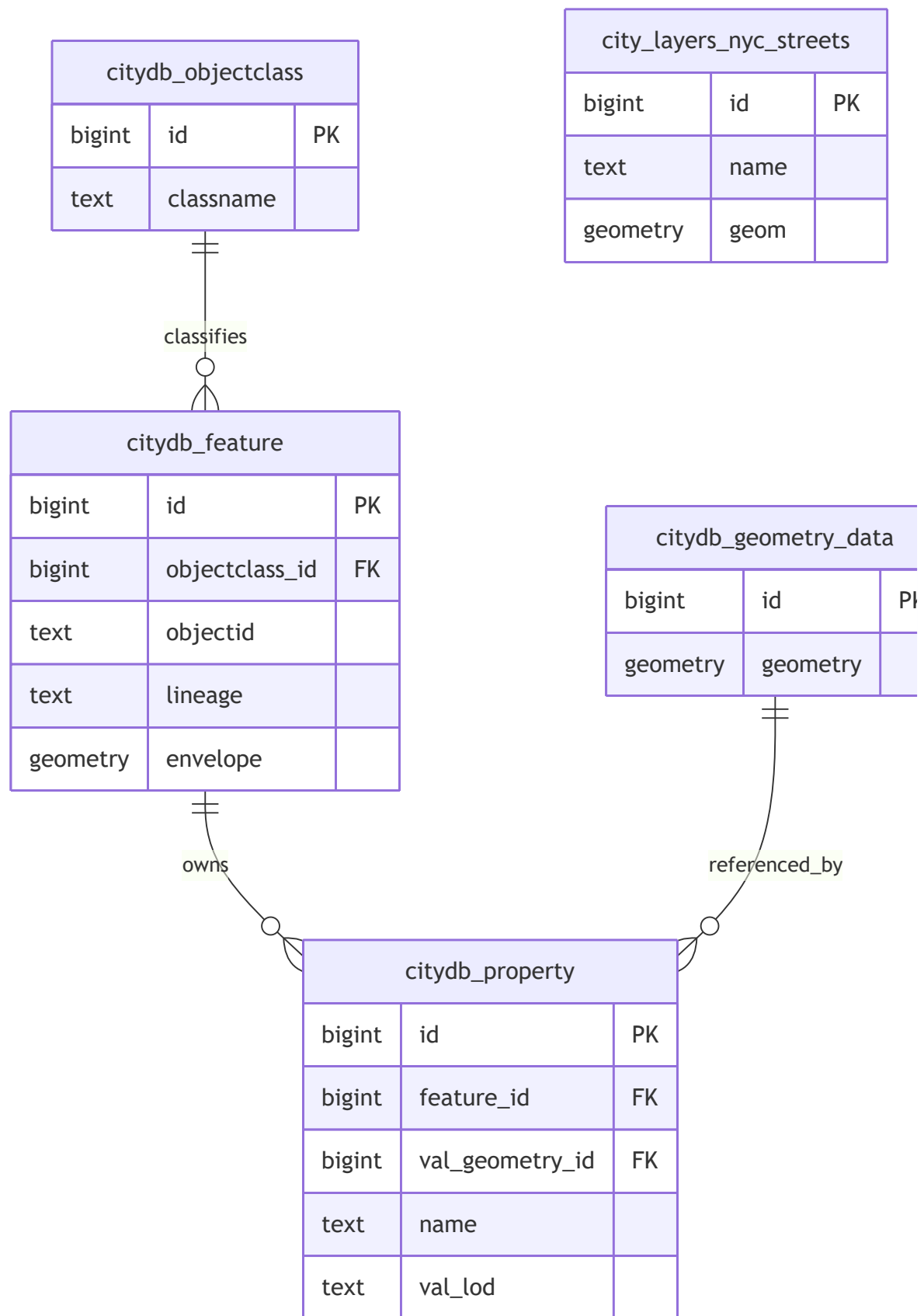
- `src/client/app/GalleryMap.tsx`
- `src/client/app/GalleryPanel.tsx`
- `src/client/stores/GalleryStore.ts`
- `src/client/stores/GalleryMapStore.ts`
- `src/client/stores/GalleryCameraController.ts`
- `src/client/stores/CameraInteractionController.ts`
- `src/client/stores/TourTransitionAnimator.ts`
- `src/client/focus/FocusController.ts`
- `src/client/focus/CameraViewController.ts`
- `src/client/routing/RoutePlanner.ts`

Server classes



The server is a Vite plugin, not a separate process. It registers middleware for `/api/citydb/*` and `/3dcitydb-client/*`, then delegates database access to **DataLoader** and binary tile creation to **DataTiler**.

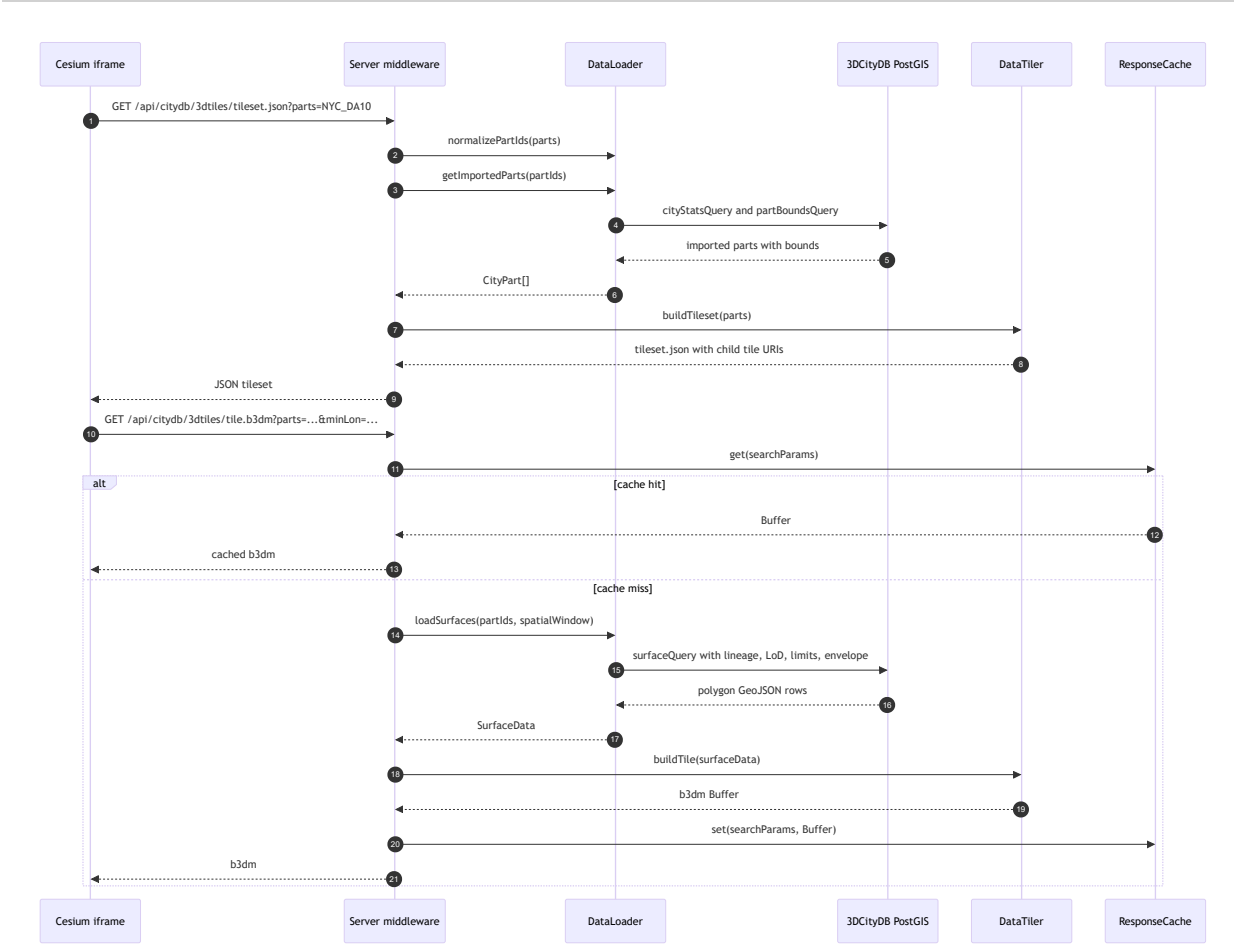
Database model used by the app



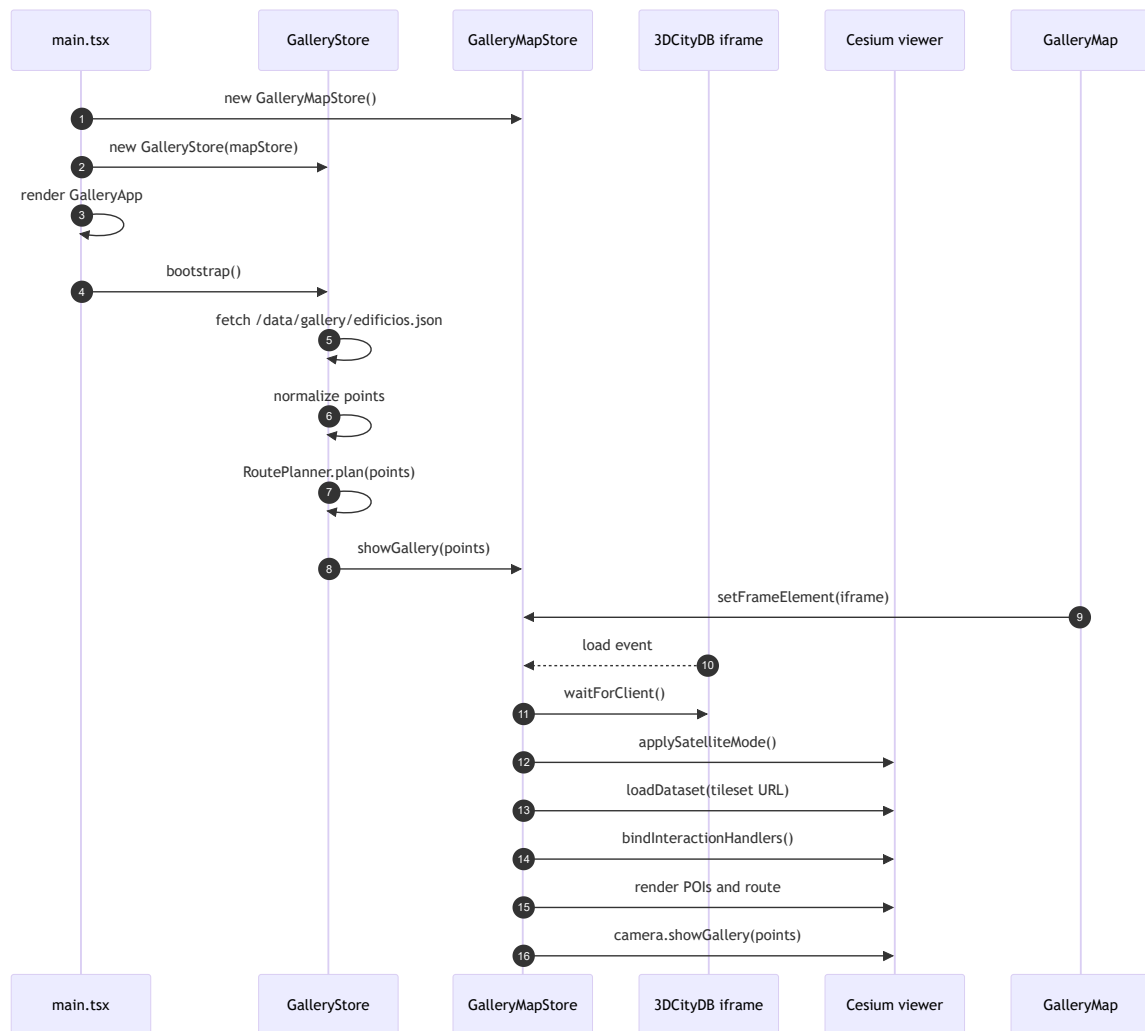
The current dynamic 3D Tiles path queries `citydb.feature`, `citydb.property`, `citydb.geometry_data`, and `citydb.objectclass`. `feature.lineage` partitions imported delivery areas such as `NYC_DA10`. `feature.envelope` provides part bounds and `geometry_data.geometry` provides renderable polygons filtered by the requested tile window.

The README also documents `city_layers.nyc_streets`, but the current `Server.ts` routes do not register the streets endpoint.

Tileset and tile request sequence

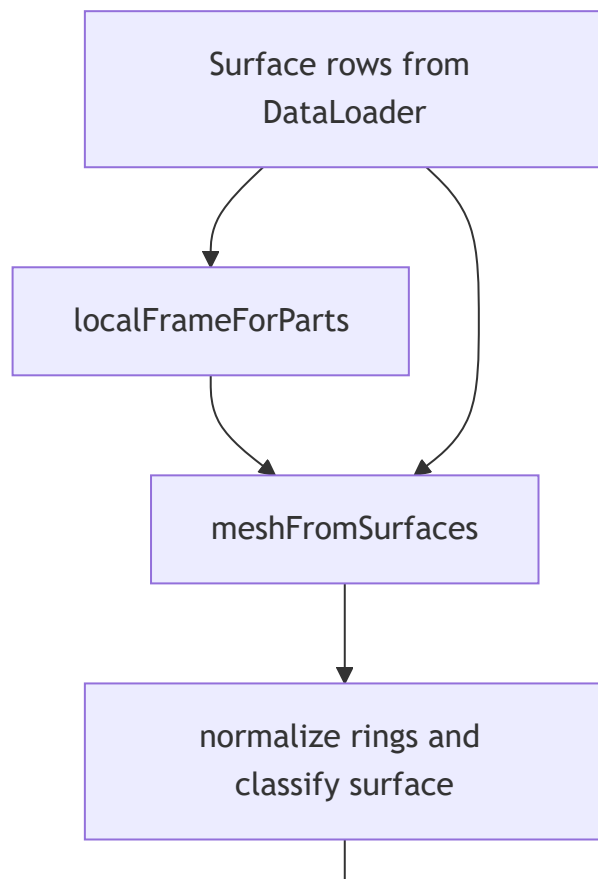
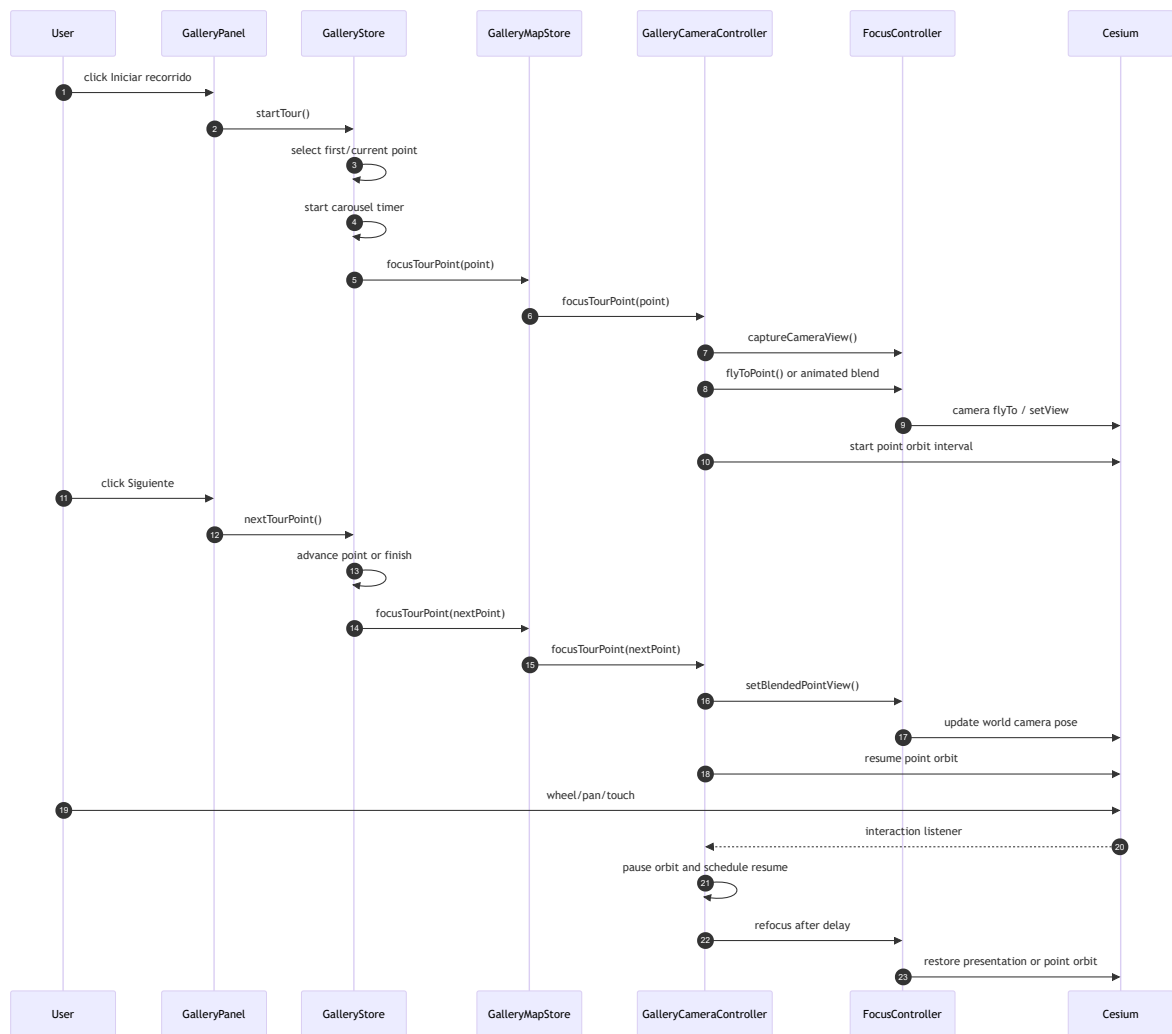


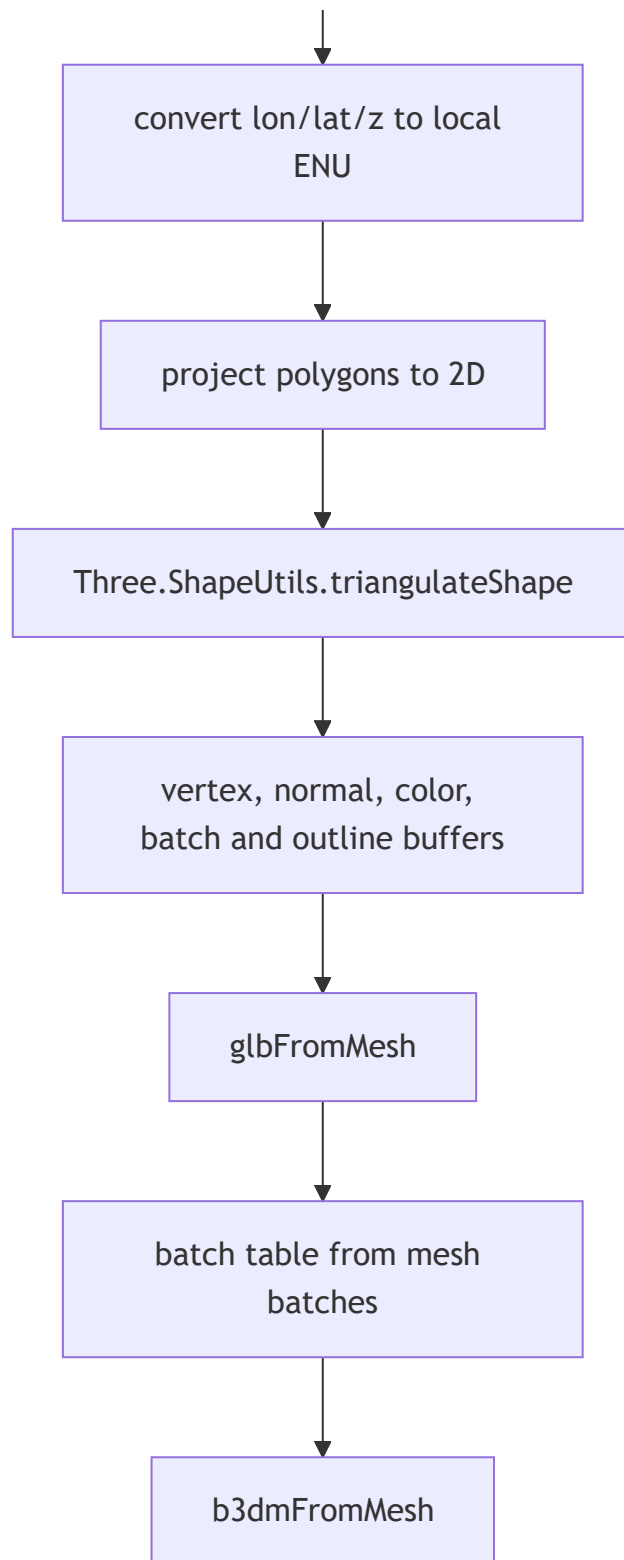
App bootstrap sequence



`GalleryStore.bootstrap()` and `GalleryMapStore.initialize()` can overlap. Both paths call `waitForClient()`, so rendering gallery entities is delayed until the iframe exposes `Cesium.Cesium3DTileset` and `cesiumViewer`.

Guided tour sequence





The generated **b3dm** contains a glTF mesh with triangle and line primitives. Batch table metadata preserves **partId**, **geometryId**, **featureId**, **objectId**, **className**, **lod**, **property**, and derived **surfaceType** for Cesium styling or inspection.